Divisibility of Reversed Consecutive-Integer Concatenations and an Arbitrary-Permutation Extension

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Abstract

For every positive integer n, we show that the concatenation of some n consecutive positive integers in decreasing order is divisible by n. Consequently, OEIS A083468 is positive and OEIS A083469 is well-defined for every positive index. We prove a more general assertion: in any even integer base, *every prescribed permutation* of n consecutive integers admits infinitely many equal-width concatenations divisible by n. The proof is constructive and rests only on congruences and the Chinese remainder theorem. We additionally determine the exact least concatenation for the infinite family $n = 10^r$ $(r \geq 1)$ by a short last-block argument, and give an explicit geometric-sum formula for the values and their quotients.

1 The two sequences

Write $u \,\|_b\, v$ for concatenation of the base-b digit strings of positive integers u, v. The sequence OEIS A083468 gives the smallest positive multiple of n obtained by concatenating n consecutive positive integers in *decreasing* order, with the value zero if no such concatenation exists. The companion sequence OEIS A083469 divides that smallest value by n; its definition does not assign a value in a hypothetical nonexistence case [1, 2]. We prove nonexistence never occurs. The proof is independent of the minimality of the listed sequence terms. It can also be viewed as a corollary of the author's even-base arithmetic-progression divisibility theorem, taking step -1 [3]. The independent proof below exposes a permutation strengthening.

2 Divisibility in every prescribed block order

Theorem 1. *Let $b \geq 2$ be an even integer, $n \geq 1$, and let $\pi : \{1, \ldots, n\} \to \{0, \ldots, n-1\}$ be an arbitrary bijection. There exist infinitely many pairs (d, k) of positive integers such that $k, k+1, \ldots, k+n-1$ all have exactly d digits in base b and*

$$n \;\Big|\; (k + \pi(1)) \,\|_b\, (k + \pi(2)) \,\|_b \cdots \|_b\, (k + \pi(n)). \tag{1}$$

In particular, there are infinitely many distinct concatenated integers satisfying (1).

Proof. Factor $n = qm$, where q is the largest divisor of n supported on primes dividing b. Thus $\gcd(q, m) = 1$ and $\gcd(m, b) = 1$. Since b is even, m is odd. Choose an arbitrarily large positive integer d with

$$q \mid b^d, \qquad b^d \equiv 1 \pmod{m}, \qquad (b-1)b^{d-1} \geq q + n - 1. \tag{2}$$

Such d exist arbitrarily far out: divisibility by q holds for all sufficiently large d; the second condition holds when d is a multiple of the multiplicative order of b modulo m (and is vacuous when $m = 1$); the inequality follows for large d. The inclusive interval

$$I_d = [b^{d-1},\, b^d - n] \cap \mathbb{Z}$$

contains at least q consecutive integers, by (2). Choose $k \in I_d$ satisfying

$$k + \pi(n) \equiv 0 \pmod{q}. \tag{3}$$

If $q = 1$, this congruence is vacuous. All n of the integers $k, \ldots, k + n - 1$ are positive and have exactly d base-b digits. Put $B = b^d$. The proposed concatenation is the integer

$$C = \sum_{i=1}^{n} \bigl(k + \pi(i)\bigr) B^{n-i}. \tag{4}$$

Since $q \mid B$, the final summand in (4) is the only one that survives modulo q, so (3) gives

$$C \equiv k + \pi(n) \equiv 0 \pmod{q}.$$

On the other hand, $B \equiv 1 \pmod{m}$, hence the permutation drops out of the residue:

$$C \equiv \sum_{i=1}^{n} (k + \pi(i)) = nk + \frac{n(n-1)}{2} \equiv 0 \pmod{m}.$$

The last congruence holds because $m \mid n$ and m is odd. As $\gcd(q, m) = 1$, the Chinese remainder theorem implies $n \mid C$. There are arbitrarily large admissible d, and the resulting concatenations have increasing lengths nd, proving infinitude. □

Corollary 1. *For every positive integer n, there exist infinitely many concatenations of n consecutive positive integers in decreasing order that are divisible by n. Consequently*

$$\boxed{\text{A083468}(n) > 0 \quad \textit{for every } n \geq 1,}$$

while $\text{A083469}(n) = \text{A083468}(n)/n$ *is well-defined and positive for every* $n \geq 1$.

Proof. In Theorem 1 take $b = 10$ and $\pi(i) = n - i$, so that the blocks occur in descending order. □

3 Exact minima at powers of the base

The existence theorem does not ordinarily identify the least concatenated integer. For power-of-base indices, however, the final block completely determines divisibility, even when earlier blocks have different widths.

Theorem 2 (Exact minimum and all admissible final blocks)**.** *Fix an arbitrary base $b \geq 2$ and an exponent $r \geq 1$, and put $N = b^r$. For a positive integer a let*

$$F_{N,b}(a) = (a + N - 1) \,\|_b\, (a + N - 2) \,\|_b \cdots \|_b\, a.$$

Then $N \mid F_{N,b}(a)$ if and only if $N \mid a$. Consequently the least *such concatenated integer is*

$$F_{N,b}(N) = (2N - 1) \,\|_b\, (2N - 2) \,\|_b \cdots \|_b\, N.$$

Proof. Write d for the number of base-b digits of a. If $d < r$, the last r base-b digits of $F_{N,b}(a)$ contain the entire final block a, which has a nonzero leading digit. Thus these r digits cannot all be zero, and N does not divide the concatenation (nor a).

If $d \geq r$, the preceding blocks together form an integer H, so

$$F_{N,b}(a) = Hb^d + a \equiv a \pmod{b^r}.$$

Therefore $N \mid F_{N,b}(a)$ exactly when $N \mid a$, in both digit-width cases.

Finally, $F_{N,b}(a)$ is strictly increasing as a function of a: increasing a increases every block, never decreases its digit length, and therefore either increases the total number of digits or increases the leading block in the fixed-width positional expansion. The smallest positive multiple of N that can serve as the final block is $a = N$, which yields the asserted exact minimum. □

Corollary 2 (OEIS A083468 and A083469)**.** *For every $r \geq 1$, write $N = 10^r$ and $B = 10^{r+1} = 10N$. Then*

$$\boxed{\text{A083468}(N) = (2N-1) \,\|_{10}\, (2N-2) \,\|_{10} \cdots \|_{10}\, N = \sum_{j=0}^{N-1} (N+j)B^j.}$$

Equivalently, this exact integer is

$$\frac{(2N-1)B^{N+1} - 2NB^N - (N-1)B + N}{(B-1)^2}.$$

Moreover,

$$\boxed{\text{A083469}(N) = \frac{1}{N} \sum_{j=0}^{N-1} (N+j)B^j.}$$

In particular, $\text{A083468}(10) = 19181716151413121110$.

Proof. Theorem 2 applies with $b = 10$. Every integer in $[N, 2N-1]$ has exactly $r+1$ decimal digits, hence descending concatenation is precisely the stated base-B positional sum. The rational expression follows from finite geometric sums, and the quotient formula is the defining relation for A083469. □

4 A concrete witness construction

For the decimal case, factor

$$n = qm, \qquad q = 2^{v_2(n)}5^{v_5(n)}, \qquad \gcd(m, 10) = 1.$$

Take an arbitrarily large d satisfying

$$q \mid 10^d, \qquad 10^d \equiv 1 \pmod{m}, \qquad 9 \cdot 10^{d-1} \geq q + n - 1.$$

The smallest multiple of q not below 10^{d-1},

$$a = q \left\lceil \frac{10^{d-1}}{q} \right\rceil,$$

lies in $[10^{d-1}, 10^d - n]$. Therefore the integer

$$(a+n-1) \,\|_{10}\, (a+n-2) \,\|_{10} \cdots \|_{10}\, a$$

is divisible by n. For instance, with $n = 4$, $d = 2$ and $a = 12$, we obtain 15141312, which is divisible by 4; this is not the minimum, which is 5432 [1].

Remark 1. *The hypothesis that b is even ensures that m is odd, which is used to annihilate $n(n-1)/2$ modulo m. No such theorem for arbitrary odd bases is claimed here.*

5 Computational checks and scope

The accompanying script `verify_reverse_concatenation.py` constructs witnesses for $1 \le n \le 512$, verifies divisibility using modular concatenation, independently reproduces the first sixteen A083468 minima, and checks the exact minimum formula at powers of ten. Finite checks are supplementary: the theorem is an unconditional proof for all indices. The general result establishes *existence* and provides explicit witnesses, but not a formula for all minimum values. The new Theorem 2 supplies exact minima and a complete classification of valid final blocks for the infinite family of power-of-base indices only. The theorem includes the previously observed reverse-progression consequence of [3]; no claim is made that the arbitrary-permutation strengthening or the special-family formula has priority over unexamined literature.

References

[1] OEIS Foundation Inc., *A083468*, https://oeis.org/A083468 (accessed October 16, 2026).

[2] OEIS Foundation Inc., *A083469*, https://oeis.org/A083469 (accessed October 16, 2026).

[3] Q. Wang, *Divisibility of Concatenated Arithmetic Progressions and Champernowne Prefixes*, https://github.com/dongxuelian2/oeis-papers/tree/main/papers/A075000-A075002, 2026. See the even-base AP divisibility theorem.

[4] F. Luca and B. Teguia Tabuguia, *Concatenations of terms of an arithmetic progression*, Journal of Difference Equations and Applications **32** (2026), 437–456, https://doi.org/10.1080/10236198.2025.2603385.